

Measurement of Neutron Multiplicity in Charged Current Neutrino Interactions on Oxygen

By

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Chapter 3

ANNIE

With the advent of neutrino beams and photodetection technologies, neutrino detectors fall into a few categories including liquid argon time projection chambers (LArTPCs), liquid scintillator (LS), and water Cherenkov (WC) detectors. In LArTPCs, like MicroBooNE and DUNE, a neutrino interaction with an argon nucleus produces a lepton that ionizes the liquid argon as it travels through the rest of the TPC. The resulting charge is collected onto wire planes and the event topology can be reconstructed from the time and position information. In liquid scintillator detectors, such as KamLAND, Daya Bay, and JUNO¹, the daughter particles deposit their energy in the scintillator, creating flashes of light that can be detected. In water Cherenkov detectors, like SNO and Super-K, the traveling lepton produces Cherenkov radiation, which is collected with photodetectors such as photomultiplier tubes (PMTs). For all detector types, large quantities of the target medium are used so that there

¹JUNO: Jiangmen Underground Neutrino Observatory

is a sufficient number of nuclear targets for neutrinos to interact with. Coupled with neutrino beam, a significant number of neutrino interaction events can be collected. We focus on the water Cherenkov detector, which has the following main advantages: (1) the relatively cheap cost of water, (2) the environmental safety of water, (3) the fast timing of the signal, (4) the ability to load neutron absorbers, and (5) the lack of natural radioisotopes.

The Accelerator Neutrino Neutron Interaction Experiment (ANNIE) is a water Cherenkov neutrino detector located on-axis in the Booster Neutrino Beam (BNB) at Fermi National Accelerator Laboratory (FNAL or Fermilab, for short). ANNIE has grown from a small experiment to a collaboration between 19 institutions and two national labs across five countries. The experiment is designed to measure the final state neutron multiplicity of neutrino-nucleus interactions, determine the charged current inclusive cross section on water, and serve as a testbed for new technologies. This chapter details the detector and the physics goals of ANNIE.

3.1 ANNIE Phase I

ANNIE is divided into several phases that mark different experimental geometries and measurement goals. The first phase of ANNIE demonstrated the feasibility of the experimental concept. The main goal was to assess the level of background neutrons and decide whether the experiment could move forward. The Phase I detector consisted of a cylindrical tank lined with white PVC, a front veto of scintillator paddles (Section [3.2.5](#)), and a partially



Figure 3.1: The Neutron Capture Volume (NCV) is an acrylic vessel filled with EJ-335, a mixture of liquid scintillator and gadolinium. A PMT coupled to the NCV is wrapped in black to prevent light generated in the tank from triggering the data acquisition system. The array of PMTs at the bottom of the Phase I detector can be seen facing upward in this image.

instrumented Muon Range Detector (Section [3.2.6](#)). At the bottom of the tank sat an array of 58 Hamamatsu R5912 photomultiplier tubes with 8-inch diameters. Above the tank lid was a set of six plastic scintillator paddles, each coupled to a 2-inch PMT via a light guide and secured to a metal frame. The front veto, MRD, and the top scintillator paddles were used to tag muons.

A large 50 cm x 50 cm acrylic cylinder filled with EJ-335, a 0.25% by weight gadolinium-loaded liquid scintillator from Eljen Technology, was deployed into the water-filled tank to measure the rate of beam-correlated background neutrons (see Figure [3.1](#)). Sources of

background neutrons include:

- **sky-shine neutrons:** neutrons that make their way out of the beam line and undergo scattering and enter the detector from above, and
- **dirt neutrons:** neutrons that are produced in the dirt.

Although not shown in Figure 3.1, this Neutron Capture Volume (NCV) was wrapped in layers of reflective material to maximize light collection and an outer layer of black plastic to optically isolate it from the rest of the tank. Two 3-inch PMTs were coupled to the NCV to trigger on scintillation light and initiate data acquisition.

The NCV was moved to different locations in the tank as indicated by the blue and red markers in Figure 3.2. After a neutron gets captured by gadolinium (Gd) in the NCV, the de-excitation photons emitted from Gd are amplified by the liquid scintillator, resulting in thousands of photons that make their way to the NCV PMTs. Their waveforms are recorded along with any activity in the tank following a beam trigger. In addition to the physics data runs, calibration runs of the NCV were carried out with a Cf-252 source. More details about the Phase I experimental setup, calibration, and data analysis can be found in [65].

Referring to Figure 3.2, we can see that an active fiducial volume placed lower in the tank (indicated by the blue curve) and bit upstream of the tank center (indicated by the red curve) sees very few background neutrons. The average beam-induced background neutron flux in this volume measures less than 0.02 neutrons/beam-spill/m³ [65]. The level of background neutrons was deemed sufficiently low to move forward with a full ANNIE detector. Since the

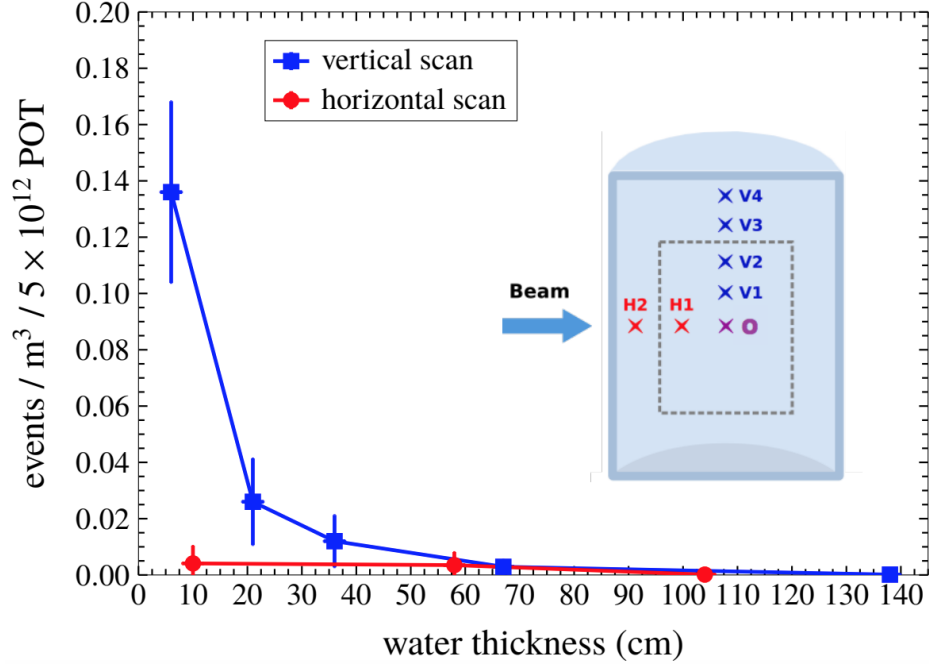


Figure 3.2: The NCV was deployed at various locations in the ANNIE Phase I detector. These background measurements show that background neutron rates are sufficiently low in a fiducial volume located in a central location slightly upstream of the tank center, indicated by the purple circle in the diagram above. Figure from [65].

Phase II installation, modifications to the area surrounding the experimental hall prompted an updated analysis of the beam-correlated neutron background. These changes include a large mound of dirt, called a berm, placed upstream of the ANNIE detector and a new experimental hall (SBND) built downstream. An updated analysis is ongoing at the time of writing this thesis.

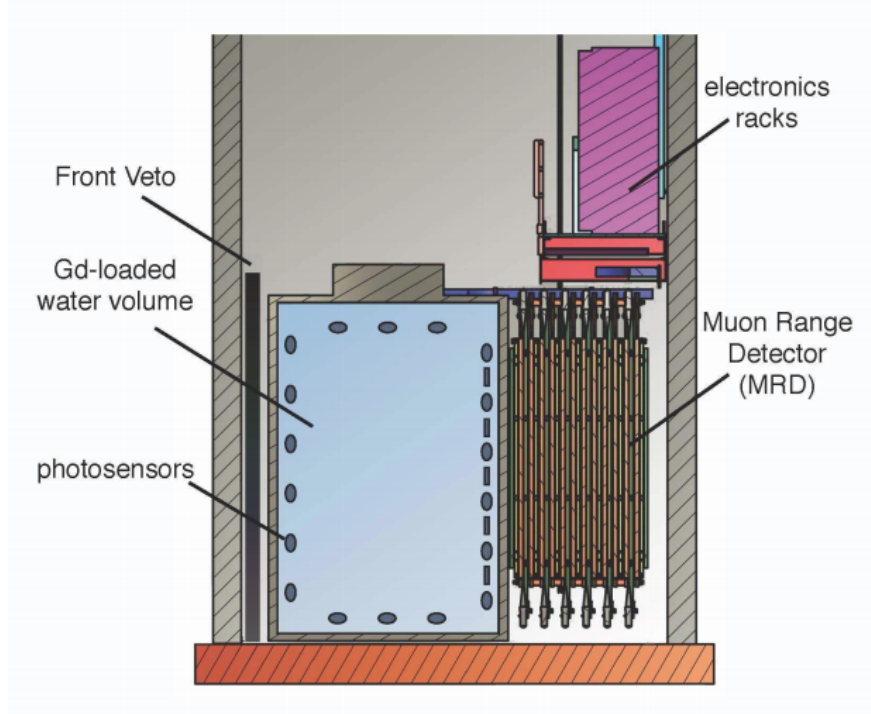


Figure 3.3: ANNIE Phase II detector diagram. The tank is 3-m in diameter and 4-m tall, with PMTs and LAPPDs lining its walls. The Front Muon Veto and Muon Range Detector sit upstream and downstream of the tank, respectively. The readout electronics and power supplies are located on the level above the detector. In this diagram, the direction of the beam is from left to right. Figure from [66].

3.2 ANNIE Phase II Detector

The second phase involved fully commissioning the ANNIE detector. Referring to Figure 3.3, the main subsystems of the Phase II detector are the

- **0.1% gadolinium-doped water target:** active medium consisting of deionized water loaded with 0.1% gadolinium by mass, giving the target an enhanced sensitivity to neutrons;

- **132 photomultiplier tubes:** conventional photomultipliers of various models and sizes used to detect Cherenkov radiation for energy and vertex reconstruction;
- **5 Large Area Picosecond Photodetectors:** novel photodetectors with picosecond timing and sub-centimeter spatial resolution for improved vertex reconstruction;
- **steel cylindrical tank and inner structure:** encapsulates the water and holds photodetectors;
- **water filtration/purification system:** network of pumps, pipes, tubes, filters, lamps, and ion exchange resins that clean and recirculate the water;
- **Front Muon Veto:** two layers of scintillator paddles for tagging charged particles created upstream of the detector;
- **Muon Range Detector:** iron-scintillator sandwich used to tag downstream muons and reconstruct their tracks;
- and **electronics rack:** system of high voltage power supplies, digitization and readout modules, and circuit boards that power the detector subsystems and take data.

The Fermilab Booster Neutrino Beam encounters the Front Muon Veto, the water tank, and Muon Range Detector in the order listed. A detailed paper describing the ANNIE Phase II detector is in the process for publication [\[67\]](#).

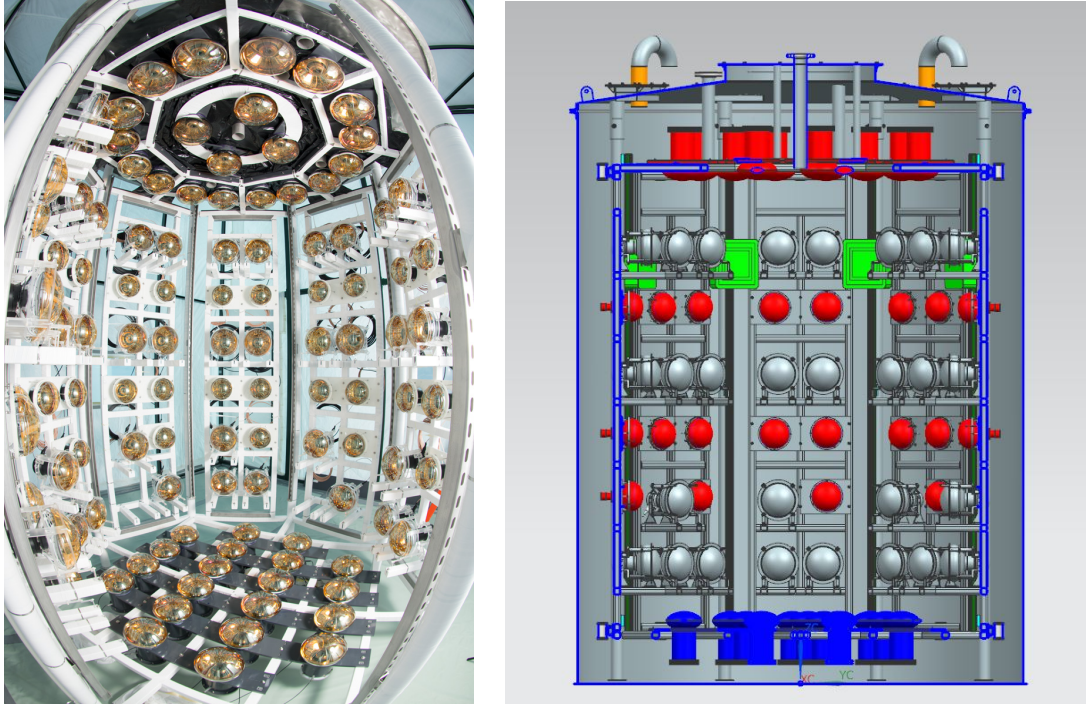


Figure 3.4: LEFT: Fishbowl perspective of the inner structure, wrapped in tape made of polytetrafluoroethylene (PTFE). RIGHT: 3D model of the inner structure. The different colors indicate different PMT types. Unistrut rails between two panels are used for LAPPD deployment.

3.2.1 Detector Tank

One of the main components of the Phase II detector is a 3-m (10-ft) diameter, 4-m (13-ft) tall right cylindrical tank constructed from 7-gauge (4.5 mm) steel, which was also used in Phase I. To prevent the formation of and contamination by rust, the inside and outside of the tank are coated with epoxy and several layers of plastic separate the water target and the steel walls of the tank. The first two layers are 6 mil polyethylene plus a 30 mil sheet at the bottom to shield the main layer from rough areas of the tank. The main layer is a

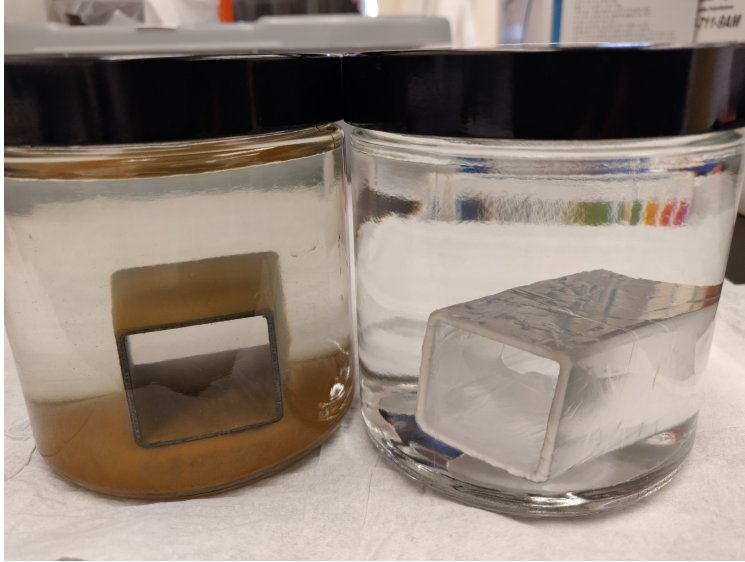


Figure 3.5: Sections of a rectangular steel pipe soaking in 1% gadolinium-loaded water. The right pipe is wrapped in teflon tape. It is clear that any presence of rust will quickly degrade the water transparency, so precautions must be taken to reduce the risk of rust contamination.

40 mil white PVC liner manufactured by Kentain [68]. The presence of rust would quickly degrade the water transparency so it is crucial that the water target does not make contact with the steel tank. Figure 3.5 compares sections of steel pipe with and without a teflon² tape wrap as a precautionary measure.

Within the liner is an octagonal stainless steel inner structure designed to hold 132 PMTs and approximately 40 LAPPDs. PMTs are mounted on one of the eight side panels, while LAPPDs are deployed along Unistrut rails that are installed where two panels meet. See Figure 3.4 for a photo and 3D representation of the inner structure. While the inner structure is made of stainless steel, the welding required to put it together compromises its resistance

²Teflon is a common brand name for polytetrafluoroethylene (PTFE)-based materials.



Figure 3.6: The inner structure is lowered into the experimental hall while photo shoots among collaborators take place. Plastic sheets protect the inner structure against dust during transport. The SBND (Short-Baseline Near Detector, as part of the Short-Baseline Neutrino Program) experimental hall is the large building to the right.

to corrosion and rust. To restore the protective passive layer, the entire inner structure was electropolished by Dr. Emrah Tiras³ and me. This electrochemical finishing process removes a thin layer of the metal with a mild acid and electrical current to repassivate its surface. As an additional preventative measure, the inner structure is wrapped with 2-inch wide PTFE tape secured with metal zip ties to slow down the rust process and leaching of contaminants. Black low-density polyethylene (LDPE) plastic sheets are wrapped around the inner structure to optically isolate the encapsulated volume. The tank lid and inner structure are seen being lowered into the hall in Figure 3.6.

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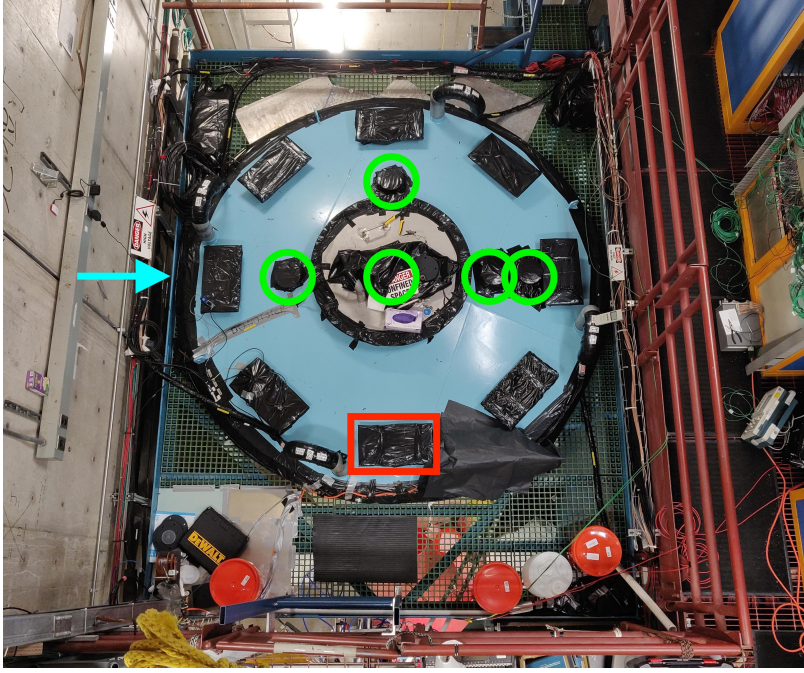


Figure 3.7: A view of the tank lid from the top level of ANNIE Hall. The five green circles indicate the locations of the ports used for source deployment. The central port is part of the larger removable hatch. The red rectangle highlights one of the eight “mail slots” used for LAPPD deployment.

The tank lid is welded to the top of the inner structure to form a single unit. The lid has several circular and rectangular ports, with a large removable hatch in the center (cf. Figure 3.7). The small circular ports permit the easy deployment and removal of calibration sources at different locations within the tank. The central hatch was designed for larger apparatuses, like the NCV of Phase I. Four PMTs are secured to the hatch, as seen in Figure 3.4, and thus, are removable. The rectangular ports, referred to as “mail slots”, are where LAPPDs will be deployed *in situ* so that commissioning of the rest of the detector can be completed in parallel with the development and testing of the LAPPD systems.

3.2.2 0.1% Gadolinium Water Target

The detector tank is filled with 26 tons of 0.1% by mass of gadolinium-loaded water. Because one of ANNIE's physics goals is to measure the rate of neutron production from neutrino interactions in the tank, gadolinium (Gd) was added to the water due to its high average neutron capture cross section of about 49,000 barns (b), compared to hydrogen's 0.33 b and oxygen's 0.19 mb [69]. Natural gadolinium consists of 15.7% Gd-155 and 14.5% Gd-157, which have capture cross sections of 61,000 b and 254,000 b, respectively. With such high cross sections, the addition of just 0.1% natural Gd by weight results in 90% of neutron captures happening on Gd, with the remaining 10% on hydrogen [70]. The time to capture is also reduced to $\sim 30 \mu s$ compared to $\sim 200 \mu s$ in pure water [71]. Any more Gd would significantly reduce the capture time constant, making it difficult to distinguish neutron captures from muons traversing through the detector.

Another benefit of gadolinium for neutron detection is the increase of visible energy in its de-excitation gammas. Following a neutron capture by Gd-155 and Gd-157, the isotopes decay through a cascade of energy levels, emitting a total of 8 MeV in energy shared among 3-4 gammas⁴ [70]. This unique signal is much more discernible than the 2 MeV gammas that hydrogen emits.

Other experiments that have used gadolinium-doped water as their neutron-tagging

⁴The peak energy is 5 MeV and ranges from 3-8 MeV as the Compton-scattered electrons eventually go below the Cherenkov threshold for detection.

medium include WATCHBOY⁵ [72] and GADZOOKS!⁶ [70]. At the time of this writing, the Super-Kamiokande experiment is beginning its Gd phase, Super-K-Gd (SK-Gd). While WATCHBOY uses GdCl_3 as its source of Gd, GADZOOKS! and ANNIE use $\text{Gd}_2(\text{SO}_4)_3$ due to the less reactive nature of SO_4^{2-} ions. The first milestone of the GADZOOKS! project is the EGADS⁷ demonstrator [73].

3.2.2.1 Cherenkov Radiation

While gadolinium is an important component, the majority of the target is pure, deionized water. When light travels through a dielectric medium such as water, its speed is reduced to $c' = c/n$, where c is the speed of light in vacuum and n is the index of refraction of the medium and $n > 1$. When this happens, it is then possible for energetic, charged particles to travel faster than light in the same medium, much like a sonic boom.

A charged particle with sufficient energy traveling at a velocity $\beta > 1/n$ through water leaves a trail of polarized molecules. Photons carry away the excess energy as the molecules relax, creating the phenomenon known as *Cherenkov radiation*, which can be detected in transparent media. The photons constructively interfere, producing a wavefront that defines a cone with an opening angle that satisfies

$$\cos \theta_C = \frac{1}{n\beta} \quad (3.1)$$

⁵A 2-ton prototype for WATCHMAN (WATER CHerenkov Monitor for AntiNeutrinos).

⁶Gadolinium Antineutrino Detector Zealously Outperforming Old Kamiokande, Super!

⁷Evaluating Gadolinium's Action on Detector Systems

Medium	Index of Refraction (n)	Cherenkov Angle (θ_C)
Air	1	$< 1.4^\circ$
Water	1.33	42°
Mineral Oil	1.47	47°
Diamond	2.42	63°

Table 3.1: Examples of media of different refractive indices n and Cherenkov angles θ_C .

For ultra-relativistic particles with $\beta \approx 1$ traveling through water ($n = 1.33$), the Cherenkov angle is constant and has the maximum value of 42° . Table 3.1 lists examples of other media used in physics.

The number of Cherenkov photons per unit distance traveled per wavelength emitted by a particle with charge Ze can be described by the Frank-Tamm formula⁸

$$\frac{d^2 N}{dx d\lambda} = \frac{2\pi\alpha Z^2}{\lambda^2} \left(1 - \frac{1}{\beta^2 n^2(\lambda)} \right) = \frac{2\pi\alpha Z^2}{\lambda^2} \sin^2 \theta_C \quad (3.2)$$

where α is the fine structure constant. The β^{-2} dependence indicates that the rate of photon emission is weakly dependent on the particle's energy, so the number of photons emitted per unit distance is roughly constant. The λ^{-2} dependence results in a greater intensity at shorter wavelengths, hence why the “blue glow” is often seen in nuclear reactor cores (cf. Figure 3.8).

For an ultra-relativistic particle of unit charge, the average number of photons emitted per cm is ~ 390 in the wavelength region of 300-700 nm. Thus, it is important that we keep

⁸Named after physicists Ilya Frank and Igor Tamm, who developed the theory in 1937 and won the 1958 Nobel Prize for their work.



Figure 3.8: The blue glow of Cherenkov radiation can be seen in the core of a nuclear reactor. This is a consequence of the phenomenon's λ^{-2} dependence, resulting in higher intensities of light at shorter wavelengths.

the transparency of the water target in this wavelength region as high as possible so that we can collect as much light as possible.

3.2.2.2 Material Compatibility

Maintaining high transparency of the water target is important if we want to detect as many Cherenkov photons as possible for accurate event reconstruction. Materials can break down or leach contaminants when submerged in solution, decreasing target transparency. As such, all materials are soaked in a 1% gadolinium (III) sulfate solution and must meet compatibility requirements before installation in the detector tank. The intent of using a

solution with a ten times higher concentration of Gd is to accelerate the appearance of any detrimental effects.

I led the efforts in the compatibility studies at UC Davis, which were later continued by others. My test procedure is as follows:

1. A batch of 1% Gd sulfate solution is prepared by dissolving 10 g of $\text{Gd}_2(\text{SO}_4)_3$ into ~ 1 L of deionized water. This is referred to as the soak solution.
2. Material samples are cleaned with ethanol and rinsed in deionized water to remove oils and dust and placed in glass jars filled with the soak solution.
3. A small volume sample of the Gd sulfate solution (no contact with any materials yet) is placed in a spectrophotometer and its UV-vis spectrum is taken. This is the control.
4. A sample of the soak solution is taken from the jar that a material is soaking in for spectrophotometry measurements.

Regular measurements were taken with a Shimadzu UV-1800 UV/Visible Spectrophotometer to monitor water transparency over the course a few weeks to several months. Materials tested include entire photomultiplier tubes, acrylic, PVC, and stainless steel.

The criteria for deciding whether a material makes the cut are that the absorbance remains nearly unchanged and no additional features show up compared to the control sample. As shown in Figure [3.9](#), the absorbance at shorter wavelengths increases over the course of a few days and weeks for some materials. However, since the photodetectors

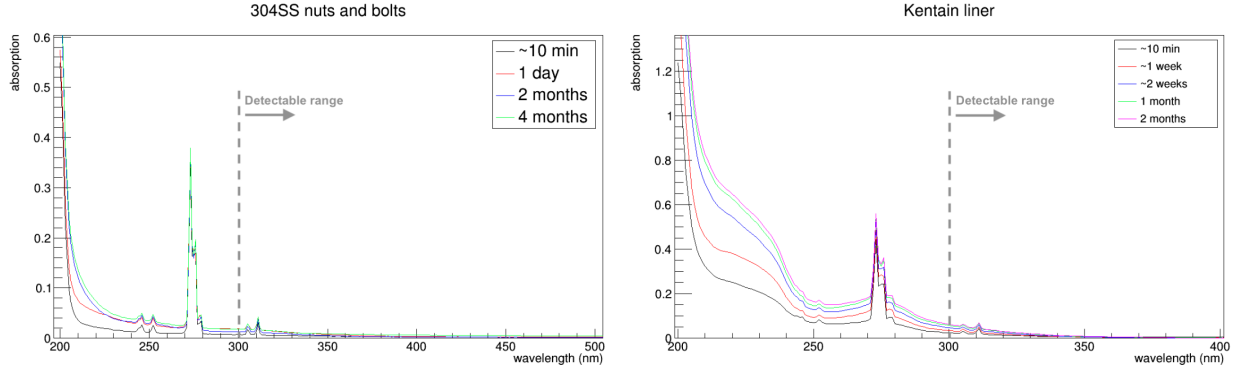


Figure 3.9: Examples of spectra for different materials soaking in 1% gadolinium-loaded water. Samples of the soak solution are analyzed frequently in the first few weeks of testing, and gradually reduce in frequency. The increasing absorbance at wavelengths below 300 nm will not affect light collection in ANNIE, but may indicate that something is leaching into the water.

are opaque to wavelengths below 300 nm, this is not a concern, though it may indicate that something is leaching into the water. To make an assessment based on the ANNIE detector, the ratio of exposed surface area of the material and the soak volume is scaled up to extrapolate absorbance and transparency values and attenuation lengths for the actual detector size.

Another factor that would disqualify a material from installation in the detector is if it physically starts to degrade. Some examples are in Figure 3.10. Clumps of material can be seen suspended in the solution, scattering light and reducing the attenuation length. Less light would be observed by the photodetectors.

The main record of materials and their properties can be found here [74].

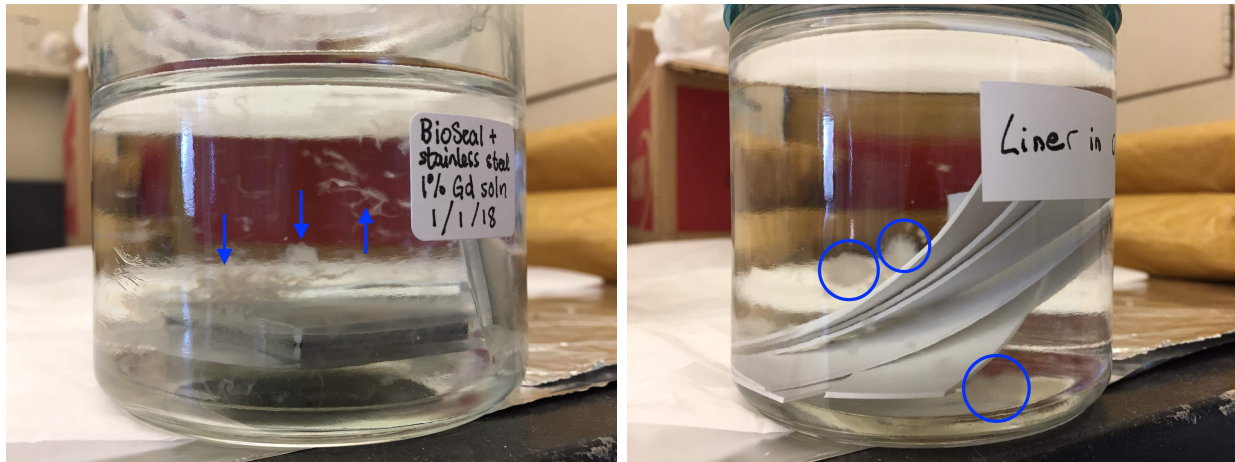


Figure 3.10: Materials degrading in gadolinium loaded water. The blue arrows and circles indicate that the material is deteriorating in the solution. This eliminates the material from use in the detector.

3.2.3 Water Filtration System

While we do our best to prevent contaminants from entering the water target, that is not enough to maintain the water transparency. Over time, dust, iron ions, and other particulates will make their way into the water target. If no actions are taken, they will accumulate in the tank, absorbing and scattering light and attenuating the signal strength by the time it reaches a photodetector. This prevents us from seeing events that otherwise would have been seen by the ANNIE detector.

In addition to electropolishing the inner structure, wrapping it with PTFE tape, and taking care to select compatible materials, the water target is circulated through a custom filtration system. The system is scaled down from the larger filtration system of the EGADS demonstrator [75]. Referring to Figure 3.11 for a diagram of the custom filtration system

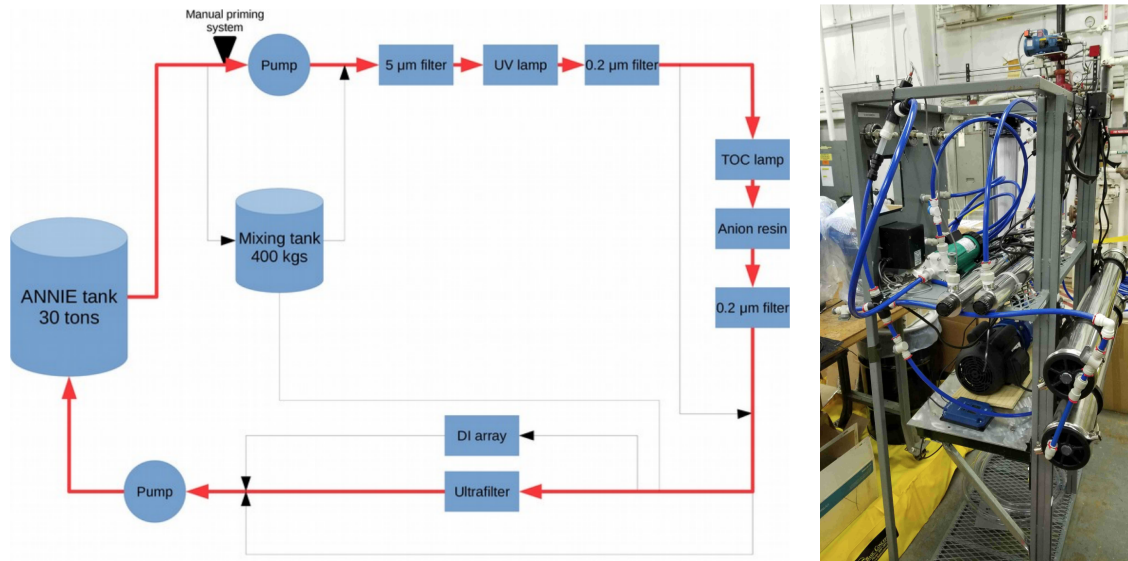


Figure 3.11: LEFT: Diagram of water flow through the ANNIE’s custom water filtration system. Black arrows mark points where the flow can be manually stopped for maintenance and testing. RIGHT: Testing the water system at Fermilab before installation in the hall. Blue tubing connect the pump, filters, ion exchanger, and lamps.

developed at UC Davis [76], the water from the main tank first encounters a 2-micron (μm) filter that is intended to remove large particles such as dust. Next the water is filtered with a ion exchange resin that removes negative ions while keeping Gd^{3+} ions in the water. This step in the filtration system removes any contaminant ions that may degrade other materials and produce more contaminants in the detector tank. Ref. [76] contains more details on selecting and producing the best ion exchange resin for ANNIE. After passing through the ion exchanger, the water then passes a 0.1-micron filter that removes even smaller particulates, such as the ion exchange beads that may come out of the ion bed. Finally, the water is sent through UV and TOC⁹ lamps that are used to stop microorganism

⁹Total Organic/Oxidizable Carbon

growth and break down any carbon compounds.

Inside the tank itself, a nitrogen blanket sits above the surface of the water to displace any oxygen molecules, further preventing microorganisms from growing and contaminating the water. Nitrogen is also circulated throughout the water.

The water system was used to load gadolinium, a process in which I was involved. The mixture was created by dissolving 52 kg of $\text{Gd}_2(\text{SO}_4)_3$ in pure water to make batches of 1% gadolinium solution. The concentrated solution is loaded into a 110-gallon mixing tank and circulated in the filtration system before it is mixed with the tank water. Following each loading is a wait period to monitor the transparency and quality of the water. When no significant changes are perceived, the next batch is loaded until 100% loading is achieved.

3.2.4 Photodetectors

ANNIE Phase II is instrumented with 132 conventional photomultiplier tubes (PMTs) and 5 novel Large Area Picosecond Photodetectors (LAPPDs), providing about 20% photocathode coverage. Of the 132 PMTs, 44 of them were purchased new from Hamamatsu, 48 are from the WATCHBOY experiment, 4 borrowed from WATCHMAN, 20 are Hamamatsu PMTs from LUX, and 20 are from Electron Tube Enterprises Ltd (ETEL). Table 3.2 lists the specifications of each PMT.

The LUX PMTs are installed in an array at the base of the inner structure. The WATCHBOY, WATCHMAN, and ANNIE PMTs are distributed throughout the side panels of the

PMT Type	Model	Diameter	QE	# in tank
ETEL	ETEL D784UKFLB	11"	25%	20
LUX	Hamamatsu R7081	10"	25%	20
WATCHBOY	Hamamatsu R7081	10"	25%	48
WATCHMAN	Hamamatsu R7081-100	10"	35%	4
ANNIE	Hamamatsu R5912-100	8"	35%	44

Table 3.2: The different types of PMT installed in the ANNIE Phase II detector. Listed for each PMT type is the PMT model, diameter, average quantum efficiency, and number installed.

inner structure. The ETEL PMTs are installed at the top, underneath the tank lid, since their ability to withstand the water pressure is uncertain.

Before their deployment in the detector tank, a portion of the total PMTs were tested at UC Davis and Fermilab to characterize their features and properties under lab settings. The WATCHBOY and ETEL PMTs were tested in a dark box at UC Davis with an LED to measure their gain values (see Chapter 5). All PMTs were tested in a dark box at Fermilab to determine or confirm gain values. More information about the lab testing of PMTs can be found in [77].

Unlike the PMTs which need to be installed before water fill, LAPPDs can be deployed at any time along one of the eight Unistrut rails located under a mail slot on the tank lid (see Figure 3.7). Chapter 4 provides a deeper discussion on LAPPDs.

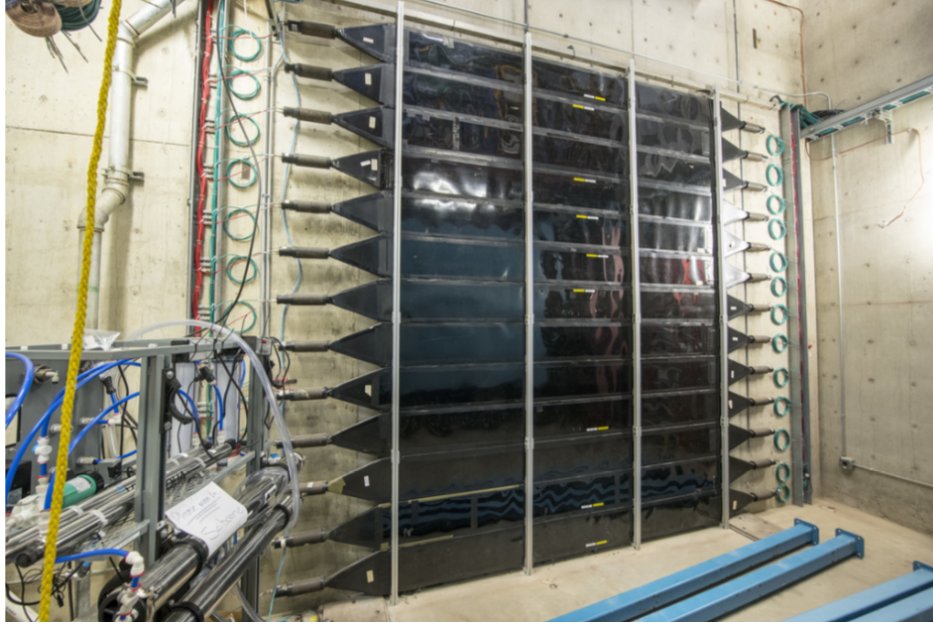


Figure 3.12: An image of the Front Muon Veto (FMV) taken from [77], before the steel tank was placed in the hall. The two layers are placed in opposite orientations and are slightly offset in height from each other to make up for the decreasing efficiency of the scintillator paddles at far end and cover any gaps between layers.

3.2.5 Front Muon Veto (FMV)

Upstream of the tank is the Front Muon Veto (FMV): two layers of plastic scintillator paddles stacked horizontally on top of each other (Figure 3.12). Each layer has 13 paddles for a total of 26 paddles, all recycled from the CDF¹⁰ experiment at Fermilab [78]. Each paddle is 322 cm long, 31 cm wide, and 2.5 cm thick, and is coupled to a 2-inch (5-cm) diameter EMI-9815 PMT via a tapered glass light guide. Typical operating voltages of the PMTs are in the 2,000-2,500 V range. The two layers are vertically offset from each other by 0.5 cm to cover any gaps between paddles. Overall, the FMV is about 3.2 m wide and 4 m tall.

¹⁰Collider Detector at Fermilab

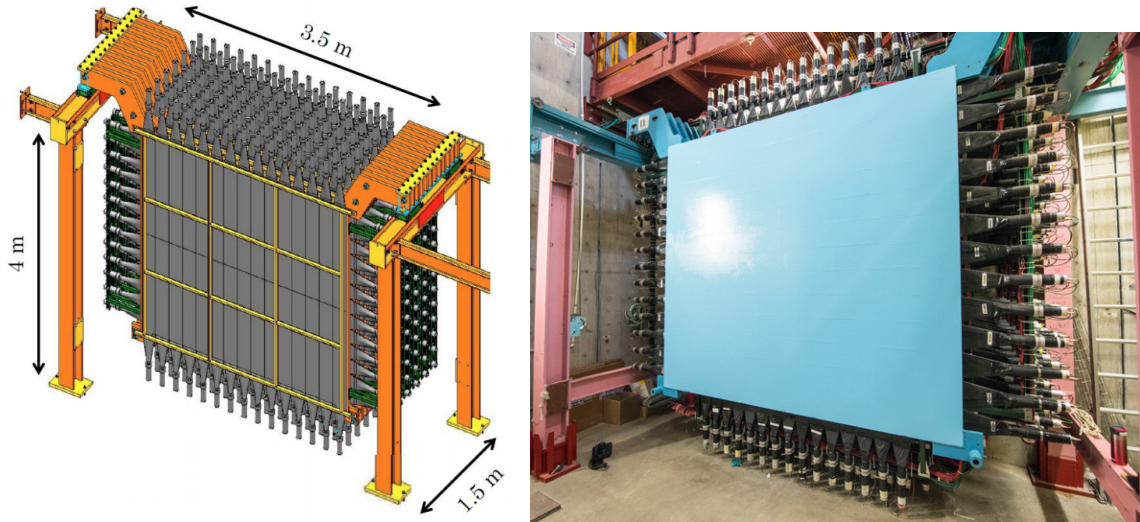


Figure 3.13: The MRD is an iron-scintillator sandwich held by an aluminum support frame. The scintillator paddles are coupled to PMTs and alternate between horizontal and vertical orientations. Diagram from [77] and picture from [67].

The FMV tag muons and other charged particles that were produced upstream and therefore did not originate in the tank. Tagging upstream muons allows their removal from the event sample.

3.2.6 Muon Range Detector (MRD)

Similar to the FMV, the Muon Range Detector (MRD) tags muons that make their way downstream of the tank. These are most muons since the ANNIE tank is relatively small, so they lose most of their energy and decay outside of the tank. The MRD is a legacy of the SciBooNE experiment [79], which ran in the same experimental hall that ANNIE now occupies. It comprises alternating layers of iron plate absorbers, which slow down the muons,

and scintillator paddles, which mark their passage through the MRD. Each iron layer has an area of 274 cm x 305 cm with a thickness of 5 cm and an average density of 7.841 g/cm³ (Figure 3.13). The scintillator layers consist of paddles that alternate in orientation and have various dimensions. The horizontally oriented paddles, all originally from SciBooNE, are 20 cm wide, 155 cm long, and 6 mm thick. The original vertical paddles are 20 cm wide, 138 cm long, and 6 mm thick. Two vertical layers were refurbished with components of the paddles from the KTeV experiment. These are 15 cm wide, 150 cm long, and 14 mm thick. The alternating orientation allows for xy -position reconstruction of the particle's track. Information from the MRD is also used to reconstruct the momentum and direction. In total, there are 306 MRD paddles spread over 6 horizontal and 5 vertical layers, with a total of 11 iron slabs and an overall dimension of 3.5 m wide and 4 m tall. The paddle efficiencies range from 80-95%. More details about the MRD in ANNIE such as PMT model and refurbishment process can be found in [67] and [77].

3.2.7 DAQ and HV

ANNIE's electronics system handles the collection of waveforms, timestamps, and so on, so that data analysis can be performed offline. It also powers the photodetectors in the tank. The FMV and MRD PMT signals are collected by Time-to-Digital Converters (TDCs), so only time information is saved when it comes to these PMTs; there is no charge information from the FMV and MRD. Their signals are discriminated by LeCroy 4413 CAMAC discrim-

inators before readout by LeCroy 3377 TDCs. High voltage for these PMTs are supplied by LeCroy 1440 HV crates.

On the other hand, readout of the tank PMT waveforms are managed by a VME system, and both the time and waveform information are saved so that charge information can be extracted. These signals are digitized by Analog-to-Digital Converter (ADC) cards and may be amplified if the PMT gain is too low to meet thresholds set at the discriminator level. The tank PMTs are powered by CAEN SY527 crates and their waveforms are recorded with VME digitizer cards originally designed for the KOTO experiment [80]. Both the CAEN and LeCroy HV crates can be remotely controlled by experts and monitored by shifters.

The integration of LAPPDs to the existing DAQ and HV systems require some modifications to the hardware and firmware, which is currently underway. Readout of the LAPPDs is performed by custom made electronics boards, which are discussed in Chapter 4.

3.2.7.1 Trigger Scheme

ANNIE has several trigger schemes for different data acquisition scenarios. Various data-taking runs include

- **Beam:** data acquisition with physics quality beam;
- **Cosmic:** data acquisition when a cosmic muon triggers a readout;
- **LED:** data acquisition with LED light sources; and
- **AmBe:** data acquisition with an AmBe run deployed in the water tank.

When the Booster Neutrino Beam is on, Fermilab's Accelerator Division sends a signal informing experiments that beam is on its way. ANNIE uses this beam signal to prepare the electronics for data acquisition. A $2\text{-}\mu\text{s}$ window is open and is extended to $70\text{-}\mu\text{s}$ if there is activity in the tank. At least one PMT with a registered hit equivalent to 5 photoelectrons is required to extend the acquisition window. These conditions make up the beam runs. Occasionally, random beam trigger windows are extended for studies on beam-correlated activity. In cosmic runs, the trigger scheme is set up to readout all subdetector systems if the first and last layers of the MRD register coincident hits from cosmic muons passing through the detector.

In LED runs, the LEDs that are installed on the inner structure are fired and a readout of the entire tank occurs. No FMV or MRD PMT data is recorded in the acquisitions. The goal of these runs is to monitor the transparency of the water. The LEDs were also used to get a first estimate of cable lengths and delays. However, a laser source was later used to better quantify timing offsets.

AmBe runs are used when an AmBe source [\[5\]](#) is deployed in the tank to characterize the neutron detection efficiency of the detector. When the PMTs coupled to the source register a hit that meet a threshold, waveforms from all the tank PMTs are recorded.

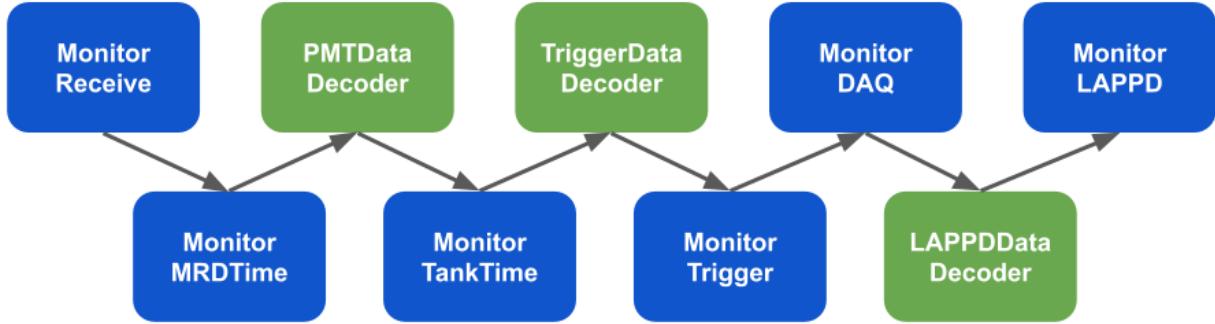


Figure 3.14: The monitoring toolchain used to monitor the subsystems of the ANNIE detector. The toolchain runs continuously to post up-to-date conditions for shifters.

3.2.8 ToolAnalysis and ToolDAQ

ToolAnalysis is a modular analysis program developed for data analysis in ANNIE. It is based on the ToolDAQ [81] framework written by Dr. Ben Richards¹¹, which is used in the DAQ systems of ANNIE and Hyper-Kamiokande [82]. Users can create tools that make up software toolchains to process and analyze data. The data and other variables are stored in `BoostStores` accessible to all tools. The diagram in Figure 3.14 depicts an example toolchain.

Each tool has an `Initialise` function that retrieves variables from configuration files and sets up input parameters, geometry, output files, and so on, that usually remain unchanged throughout the analysis chain. After initialization, any calculations and analysis techniques that need to be performed on multiple events are placed under the `Execute` function. This function is looped over for each tool until all entries are processed. Finally, the `Finalise`

¹¹University of Warwick

function closes out files, clears variables from memory, and writes information to file if the user specifies.

ANNIE’s monitoring system makes up one of the toolchains used. Monitoring tools running in a continuous tool chain produce and post plots to monitoring webpages for shifters to identify when things go wrong. More information about monitoring can be found in Chapter 5 on calibration of the ANNIE detector. Other toolchains are in charge of readout and control of other components of the DAQ system.

3.3 Booster Neutrino Beam (BNB)

The Booster Neutrino Beam (BNB) is one of the active beamlines at the Fermi National Accelerator Laboratory (Fermilab) in Batavia, Illinois. It is the primary source of neutrinos for current and past experiments such as ANNIE, SBND, ICARUS, SciBooNE, MiniBooNE, and MicroBooNE. The other neutrino beamline is the Neutrinos from the Main Injector (NuMI) beam. NuMI delivers neutrinos to the MINERvA, MINOS, and NO ν A experiments.

For the BNB to provide a steady beam of muon neutrinos, several key steps are required. First, a source of protons is created with the *Proton Source*, which consists of the Pre-Accelerator, Linac, and Booster (see Figure 3.15). Two hydrogen (H^-) ion beams of 35 keV are focused, chopped¹², and accelerated to 750 keV with magnets and radio-frequency quadrupoles (RFQs) of the Pre-Accelerator. The Linac segment of the Proton Source then

¹²An Einzel lens is used as a beam chopper. Beam chopping provides more control over the length and refines portions of the beam pulse that move on to the next stages of the accelerator complex.

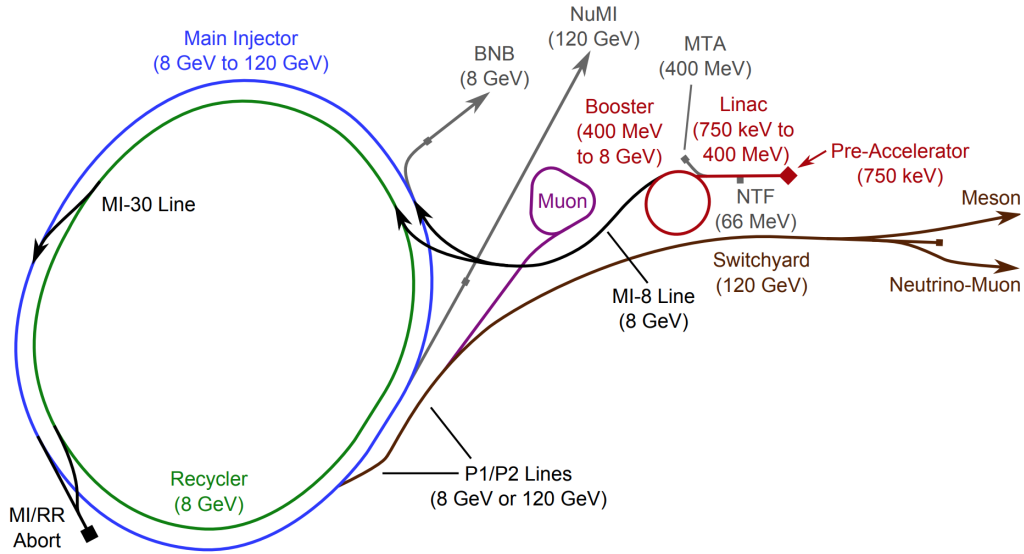


Figure 3.15: A schematic of Fermilab accelerators. The Pre-Accelerator, Linac, and Booster make up the Proton Source, which converts a beam of H^- ions into an 8 GeV proton beam. The protons are sent to several destinations, one of them to make the Booster Neutrino Beam.

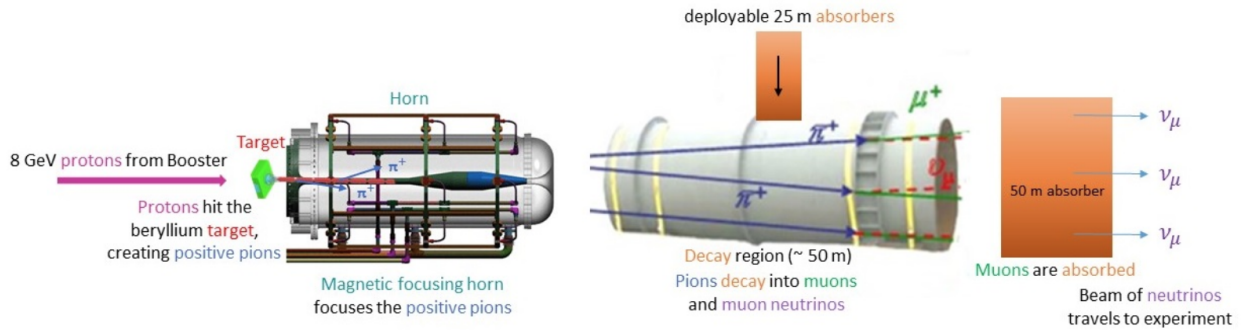


Figure 3.16: Diagram of the Booster Neutrino Beam target system at Fermilab. The Booster delivers an 8 GeV beam of protons to a beryllium target, creating pions and kaons that are focused with a magnetic focusing horn. The mesons decay into muons and muon neutrinos. The muons are absorbed and what is left is a muon (anti)neutrino beam.

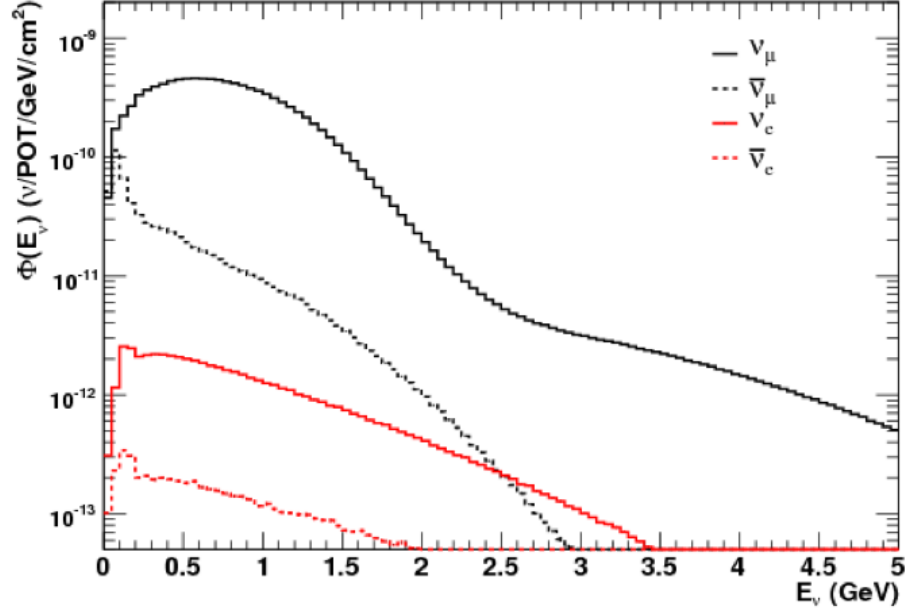


Figure 3.17: The Booster Neutrino Beam (BNB) energy spectrum shows that peak neutrino production occurs around 700 MeV. Figure from [83].

accelerates the H^- ion beam to 116.5 MeV in the low energy section and to 400 MeV in the high energy section. The 400 MeV beam is then directed to several destinations, one of them being the Booster, a 74.5 m radius (474.2 m circumference), rapid-cycling synchrotron. Before injection into this final stage of the Proton Source, the H^- ions are passed through a stripping foil to remove the electrons, resulting in a beam of protons. The Booster accelerator, using a series of magnets and RF cavity stations, focuses and boosts the 400 MeV protons to 8 GeV. The resulting beam has 84 proton bunches and a revolution period of 1.6 μs . The MI-8 Line delivers the proton beam from Proton Source to the Booster dump, the Main Injector (MI), the Recycler, or the BNB beamline.

To create the BNB, the protons are directed at a beryllium (Be) target, creating positive and negative pions and kaons. The pions decay into muons and neutrinos. The kaons would also decay into muons and pions, which in turn decay into more muons and neutrinos. The mesons are sent through an electromagnetic focusing horn, originally built for MicroBooNE, where magnets are used to select the desired charge, or “sign”, of the pions. Flipping the magnetic field produces a neutrino beam of the opposite sign. The beam of pions is collimated about 259 cm downstream of the Be target and sent to a decay region of about 50 m in length, where they can decay into muons and muon neutrinos. A steel and concrete absorber terminates the decay region, stopping the muons while the muon neutrinos continue forward. The resulting neutrino beam is about 93% ν_μ and 6.4% $\bar{\nu}_\mu$ from $p + Be \rightarrow \pi^- \rightarrow \mu^- + \bar{\nu}_\mu$ and 0.6% ν_e from decays of μ^+ produced in the main π^+ decay channel and other channels. More details about the accelerator, BNB, and other beamlines at Fermilab can be found in [84] and [85].

ANNIE sits about 100 m downstream from the beryllium target, placing it in a on-axis of the BNB. About 5×10^{12} protons-on-target (POT) per spill are delivered to the BNB with an average rate of 5 Hz. With these numbers, about 10,000 charged-current events per m^3 per year in the ANNIE detector [66]. ANNIE is expected to see $\mathcal{O}(10^{20})$ POT per year. This corresponds to a flux of $\sim 5.19 \cdot 10^{-10} \nu/\text{cm}^2/\text{POT}$ [83]. The flux of this beam peaks at about 700 MeV (Figure 3.17), which is in the energy region of interest of long baseline and atmospheric neutrino experiments. The close proximity to the target provides a high flux of neutrinos, which means more statistics!

Monitoring the beam conditions is necessary for precise physics measurements. When the accelerator is not producing physics quality beam, then very few physics events can be studied. Four systems are in place upstream of the target to monitor the BNB: two toroids (TOR875, TOR860) that measure beam intensity (protons-per-pulse), beam position monitors (BPM) and a multiwire chamber to determine the beam width and position, and a resistive wall monitor (RWM) to measure the time and intensity of beam spills. Measurements from these beam monitoring devices are used to identify periods of acceptable beam quality for data analysis.

3.4 Physics Program

ANNIE's physics goals are to measure the final state neutron multiplicity of neutrino interactions in water and the charged current (CC) inclusive neutrino cross section. These measurements will benefit next generation, long-baseline neutrino experiments that are studying neutrino oscillation, the Diffuse Supernova Neutrino Background (DSNB), and proton decay (PDK), by characterizing relevant backgrounds and reducing systematic uncertainties.

The ANNIE detector also functions as a testbed to demonstrate new technologies. For example, as will be discussed in a later chapter, ANNIE is the first physics experiment on a neutrino beam to deploy LAPPDs. By demonstrating the capabilities of this picosecond technology, production of these photodetectors can scale up, reducing the cost of one unit. Other experiments can also take advantage of these devices, along with the custom electronics

developed for the LAPPD system. ANNIE will demonstrate that even with just a few LAPPDs, experimental measurements can be improved significantly. With LAPPDs, ANNIE may also demonstrate multi-track reconstruction, distinguish between overlapping events, and better resolve the structure of electromagnetic showers.

ANNIE is also a testbed for new detection media: gadolinium-doped water and water-based liquid scintillator (WbLS). It is also the first Gd-doped water Cherenkov detector on a neutrino beam. Being able to identify neutrons more easily complements experiments that are unable to see neutrons but can see protons. Combining Gd-doped water and WbLS allows for the separation of Cherenkov radiation (to detect the muon) and scintillation light (to detect the proton), so that we can better reconstruct neutrino events. Coupled with the picosecond timing capabilities of the LAPPD, ANNIE will be able to demonstrate the power of these new technologies.

3.4.1 Neutron Multiplicity Measurement

Recall in section [2.4](#) that it is important to quantify the number of neutrons that are produced in neutrino-nucleus interactions because they can take away a significant fraction of the energy from the outgoing lepton, thus biasing event reconstruction. Existing neutron multiplicity measurements carried out by the SNO and Super-K experiments (see Figure [3.18](#)) were made with atmospheric neutrinos, which are a mixture of all flavors and a significant fraction of antineutrinos. The direction and sign of the neutrinos and antineutrinos were

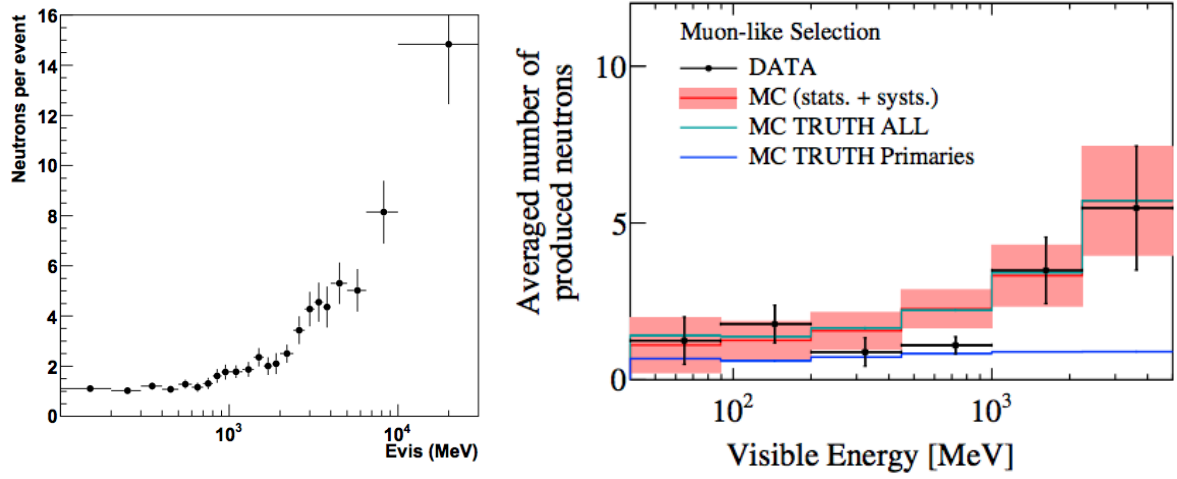


Figure 3.18: Neutron multiplicity measurements by Super-K [86] (left) and SNO [87] (right) experiments. Measurements were made with atmospheric neutrinos, whose sign and direction are not well-known. In addition, neutron yields are characterized as a function of visible energy.

also not known. ANNIE will improve upon this measurement due to its unique position. For one, ANNIE sits on a neutrino beam, so the statistics are significantly larger, the energy of the beam is constrained to a narrower range, and the neutrino flavor composition is purer (single flavor and sign). Being on a neutrino beam also gives us the knowledge of the neutrino direction. Secondly, the Super-K and SNO measurements of neutron multiplicity by were made as a function of visible energy. This can be made more precise by presenting the measurement as a function of muon momentum, or even neutrino energy. Finally, Super-K's and SNO's measurements were performed in pure water and heavy water, respectively, where the neutron detection efficiency is very low. ANNIE (and SK-Gd) has a much improved neutron detection efficiency due to its Gd loading.

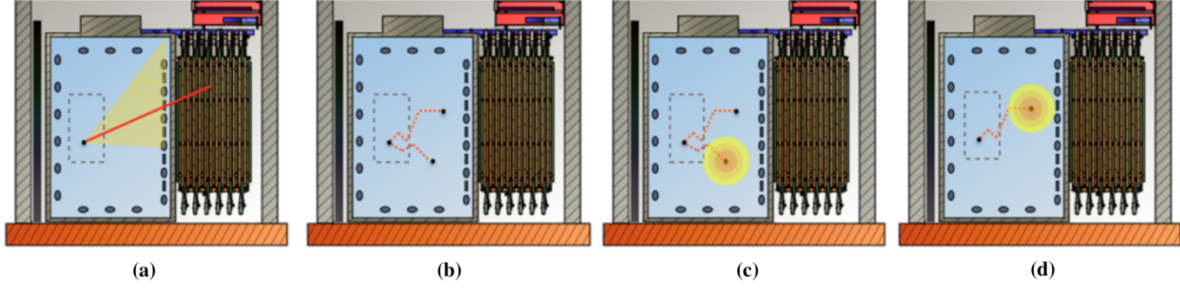


Figure 3.19: A typical event in ANNIE. (a) A neutrino-nucleus interaction creates a muon that undergoes Cherenkov radiation as it travels out of the water tank and ranges out in the MRD. (b) Final state neutrinos are created and thermalize (few μs). (c)-(d) Neutrons capture on gadolinium and produces flashes of light to be observed by photodetectors.

The neutrino interactions of interest are:

$$\begin{aligned}\nu_{\mu} + n &\rightarrow \mu^{-} + p \\ \bar{\nu}_{\mu} + p &\rightarrow \mu^{+} + n\end{aligned}\tag{3.3}$$

Since the BNB is 97% ν_{μ} , the former reaction will make up the majority of events. However, there is some contamination from $\bar{\nu}_{\mu}$ so the latter reaction will also take place in the detector. The daughter muon travels out of the detector and may stop in the MRD. A typical ANNIE event is shown in Figure [3.19](#). First, a muon neutrino interacts with a nucleus and a muon is created. The muon emits Cherenkov radiation as it exits the water tank and ranges out in the MRD, providing a prompt signal. Any final state neutrons resulting from nuclear processes thermalize until they are captured by gadolinium. The de-excitation of gadolinium produces a delayed signal in the tank.

To reconstruct the muon energy, charge and timing information from the detector tank

along with track information from the MRD are used to reconstruct the muon vertex and direction. Using simple kinematics, the Q^2 value of the neutrino interaction can be extracted. The number of neutrons can be determined by identifying clusters of hits that fall into the delayed/extended window, and can be plotted against the muon's momentum transfer. While various final state topologies exist for neutrino-nucleus interactions, this thesis focuses on the CC- 0π event topology (CC events with no pions) so that a simple kinematic model can be used to reconstruct the neutrino energy from lepton kinematics. Future studies will look at CC- 1π (CC events with one pion), NC, and other event topologies.

3.4.2 Cross Section Measurement

As mentioned in the previous chapter, oxygen has very few cross section measurements despite its common use as a target in water Cherenkov neutrino detectors. A precise measurement of the CC inclusive cross section in water is beneficial for the next generation of neutrino oscillation experiments. These experiments require low systematic uncertainties to be able to achieve their physics goals. ANNIE is in a good position to improve the cross section measurements in water due to its high statistics of neutrino interaction events. In addition, since the SBN¹³ near detector is downstream of the ANNIE hall, there is an opportunity to perform a joint argon and oxygen cross section analysis and a combined proton and neutron multiplicity measurement. Although efforts to perform this measurement are ongoing, this thesis will focus on the CC- 0π neutron multiplicity measurement.

¹³Short-Baseline Neutrino Program

3.4.3 Water-based Liquid Scintillator Technology

There is growing interest in combining the capabilities of water Cherenkov detectors with the capabilities of liquid scintillators, and being able to detect and separate scintillation light and Cherenkov radiation. The motivation for this is the fact that scintillation light can provide additional calorimetric information to improve energy resolution and the ability to detect neutral particles with energies that don't meet the Cherenkov threshold. Various methods such as spectral photon sorting, time separation with fast photodetectors, and development of novel detection medium (e.g., slow scintillators) were explored. ANNIE would use fast photodetectors, namely the LAPPDs, in combination with a WbLS medium to demonstrate this capability.

In fact, ANNIE has already done so. Details of this technical demonstration can be found in [88]. In short, ANNIE deployed a small acrylic vessel filled with WbLS in a program called Scintillator in ANNIE for Neutrino Detection Improvement (SANDI). The demonstration was successful and the tools developed to distinguish scintillation from Cherenkov radiation create a basis on which to build and develop vertex reconstruction and particle identification tools.

As we can see, there are many physics opportunities with ANNIE. The rest of this work focuses on the development and testing of LAPPDs and a $\text{CC-}0\pi$ neutron multiplicity measurement.